

**SKILLING-ML/DL****II-II CSE, CSE(AI&ML)****AY: 2025-2026**

SL No	Week	Slot-1 (3 Hours)	Detailed Subtopics	Slot-2 (3 Hours)	Detailed Subtopics
1	Week 1	Introduction to Machine Learning	Definition and scope of ML, AI vs ML vs DL, Evolution of ML, Types of learning (Supervised, Unsupervised, Semi-supervised, Reinforcement), Real-world ML applications in healthcare, finance, vision	Exploratory Data Analysis (EDA)	Dataset understanding, Data types, Descriptive statistics, Missing value handling, Outlier detection, Correlation analysis, Data visualization techniques

WEEK 1 – SLOT 2 : EXPLORATORY DATA ANALYSIS (EDA)**Duration:** 3 Hours (Theory + Hands-on)

1. Introduction to Exploratory Data Analysis
2. Role of EDA in the Machine Learning Pipeline
3. Types of Data Used in EDA
4. Descriptive Statistics
5. Data Quality Issues
 - Missing Values
 - Outliers
6. Data Visualization Techniques
7. Correlation Analysis
8. Tools for EDA in Python
9. Practical EDA Workflow
10. Summary

1. Introduction to Exploratory Data Analysis (EDA)

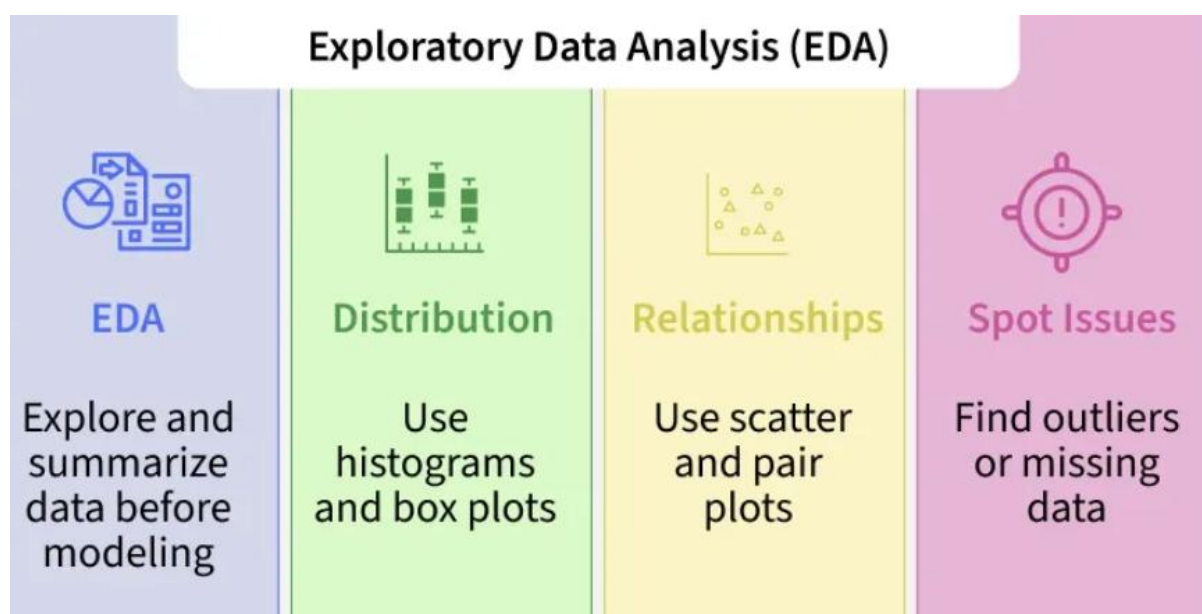
Exploratory Data Analysis (EDA) is the process of analyzing and understanding a dataset before applying any Machine Learning algorithms. It helps in summarizing the main characteristics of data using statistical measures and visual methods.

EDA allows analysts to explore the structure of data, identify patterns, detect anomalies, and uncover hidden relationships between variables. It plays a crucial role in making informed decisions about data preprocessing, feature selection, and model choice.

EDA is usually the **first analytical step after data collection and cleaning**.

Key objectives of EDA:

- Understand the nature and structure of data
- Detect missing or incorrect values
- Identify outliers and unusual patterns
- Discover relationships between variables
- Prepare data for modelling.



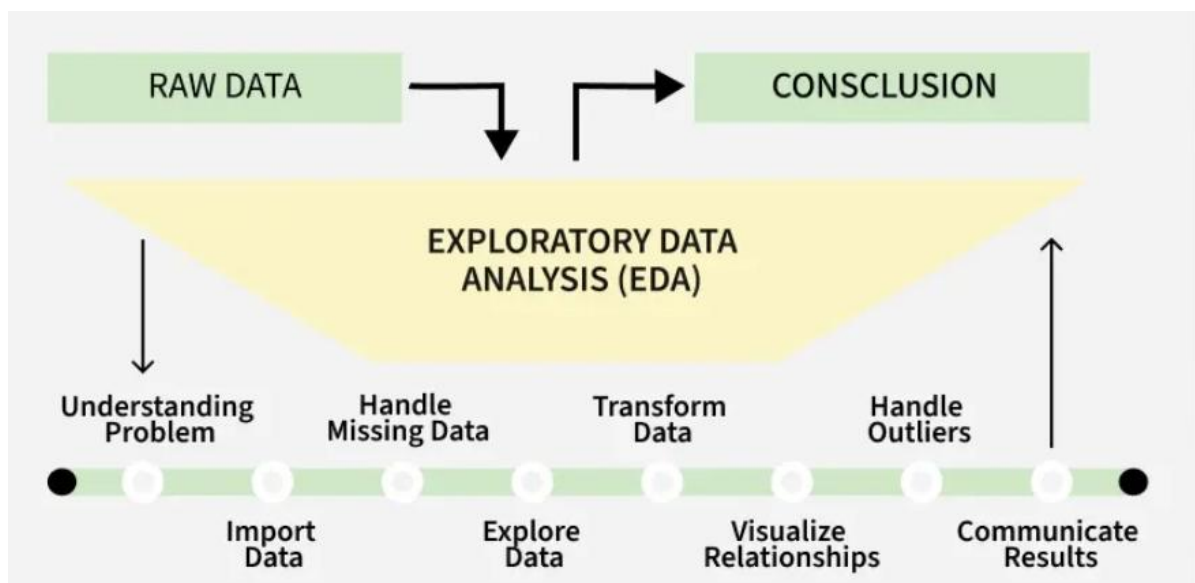
2. Role of EDA in the Machine Learning Pipeline

In the Machine Learning pipeline, EDA acts as a bridge between raw data and model building. Without EDA, models may be trained on incorrect, biased, or noisy data, leading to unreliable predictions.

EDA helps in:

- Choosing suitable Machine Learning algorithms
- Identifying important features
- Reducing dimensionality
- Preventing overfitting and underfitting
- Improving model interpretability

For example, EDA can reveal whether a dataset is linearly separable or highly skewed, which directly influences the selection of algorithms.



3. Types of Data Used in EDA

Understanding data types is essential in EDA because different data types require different statistical techniques and visualization methods.

3.1 Numerical Data

Numerical data consists of measurable quantities expressed in numbers.

- **Continuous Data:** Can take any value within a range
Example: temperature, height, weight
- **Discrete Data:** Takes specific, countable values
Example: number of students, number of transactions

3.2 Categorical Data

Categorical data represents labels or categories.

- **Nominal Data:** No inherent order
Example: gender, blood group
- **Ordinal Data:** Has a meaningful order
Example: education level, ratings

3.3 Structured and Unstructured Data

- **Structured Data:** Organized in rows and columns
Example: Excel sheets, databases
- **Unstructured Data:** No predefined format
Example: text, images, audio

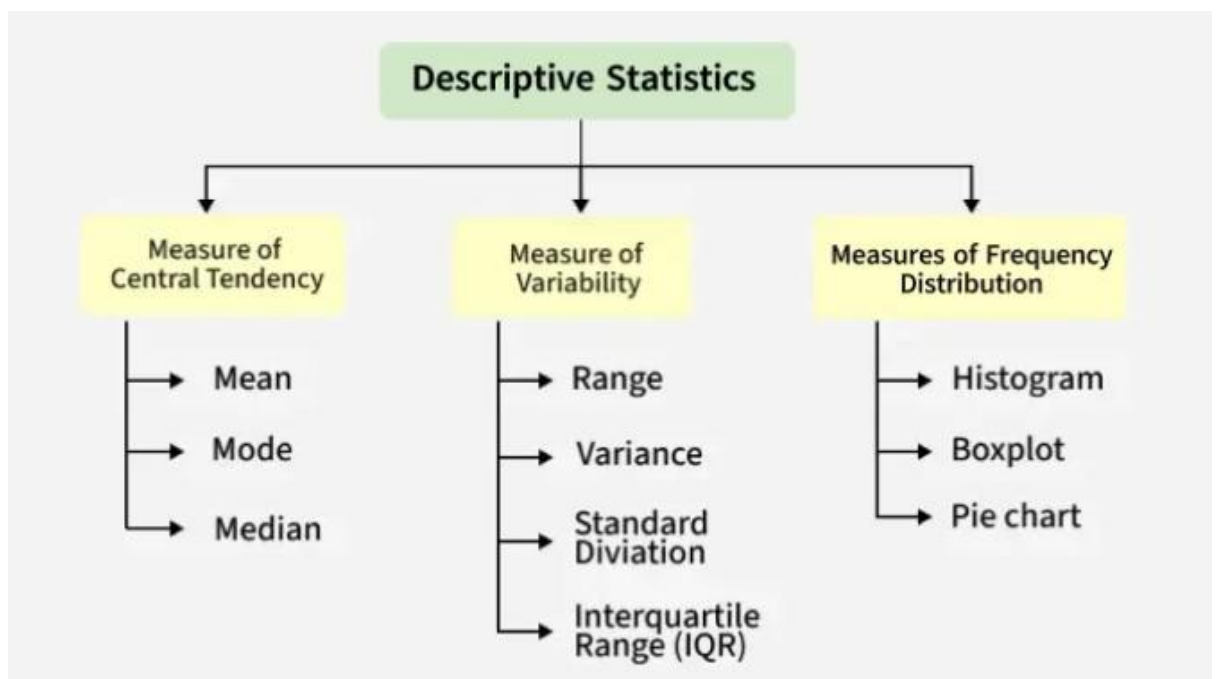
4. Descriptive Statistics

Descriptive statistics summarize and describe the main features of a dataset. They provide numerical insights into the distribution, central tendency, and variability of data.

Types of Descriptive Statistics

There are three categories for standard classification of descriptive statistics methods, each serving different purposes in summarizing and describing data. They help us understand:

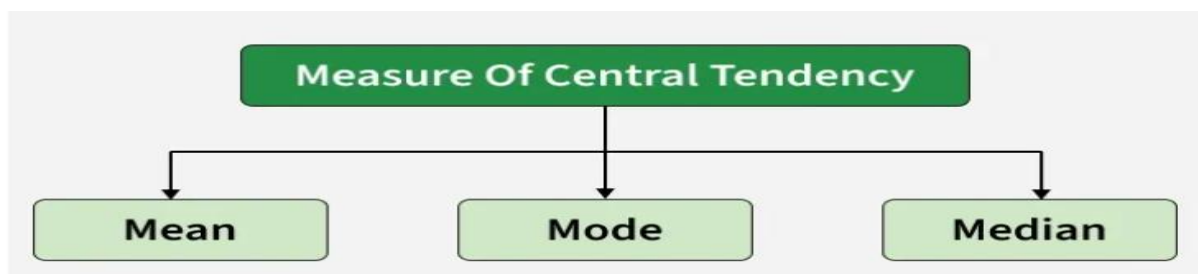
1. Where the data centers (Measures of Central Tendency)
2. How spread out the data is (Measure of Variability)
3. How the data is distributed (Measures of Frequency Distribution)



4.1 Measures of Central Tendency

- **Mean:** Average of all values
- **Median:** Middle value when data is ordered
- **Mode:** Most frequent value

The median is preferred when data contains outliers.



Mean: is the sum of observations divided by the total number of observations. It is also defined as average which is the sum divided by count.

$$\bar{x} = \frac{\sum x}{n}$$

where,

- x = Observations
- n = number of terms

Mode: The most frequently occurring value in the dataset. It's useful for categorical data and in cases where knowing the most common choice is crucial.

Median: The median is the middle value in a sorted dataset. If the number of values is odd, it's the center value, if even, it's the average of the two middle values. It's often better than the mean for skewed data.

4.2 Measure of Variability

Knowing not just where the data centers but also how it spreads out is important. Measures of variability, also called measures of dispersion, help us spot the spread or distribution of observations in a dataset. They identifying outliers, assessing model assumptions and understanding data variability in relation to its mean.

- **Range:** Difference between maximum and minimum
- **Variance:** Average squared deviation from the mean
- **Standard Deviation:** Square root of variance, indicating data spread

Lower dispersion indicates data consistency.

1. **Range :** describes the difference between the largest and smallest data point in our data set. The bigger the range, the more the spread of data and vice versa. While easy to compute range is sensitive to outliers. This measure can provide a quick sense of the data spread but should be complemented with other statistics.

Range = Largest data value - smallest data value

2. **Variance:** is defined as an average squared deviation from the mean. It is calculated by finding the difference between every data point and the average which is also known as the mean, squaring them, adding all of them and then dividing by the number of data points present in our data set.

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

where,

- $x \rightarrow$ Observation under consideration

- N -> number of terms
- μ -> Mean

3. Standard deviation: Standard deviation is widely used to measure the extent of variation or dispersion in data. It's especially important when assessing model performance (e.g., residuals) or comparing datasets with different means.

It is defined as the square root of the variance. It is calculated by finding the mean, then subtracting each number from the mean which is also known as the average and squaring the result. Adding all the values and then dividing by the no of terms followed by the square root.

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

where,

- x = Observation under consideration
- N = number of terms
- μ = Mean

4.3 Measures of Distribution

A frequency distribution table is a summarized way to show how data points are distributed across different categories or intervals. It helps to identify patterns, outliers and the overall structure of the dataset. It's often the first step in understanding the dataset before applying more advanced analytical methods or creating visualizations like histograms or pie charts.

Frequency Distribution Table Includes measure like:

- Data intervals or categories
- Frequency counts
- Relative frequencies (percentages)
- Cumulative frequencies when needed

A frequency distribution is a method for organizing data and determining how often each value occurs.

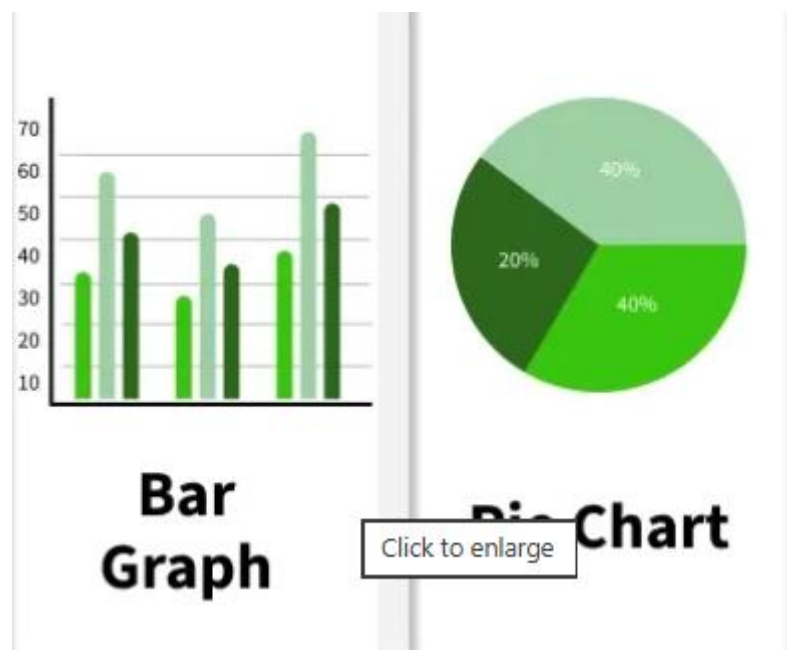
- It shows how many times each value or range of values occurs in a dataset.
- Instead of examining raw, scattered numbers, this approach presents data in a structured table or graph, making patterns, trends, and comparisons easy to identify.

Frequency distributions allow even nontechnical readers, such as founders or investors, to interpret large datasets efficiently. Today, they are considered a core component of data analysis, statistics, and data visualization.

Frequency Distribution Graphs

To represent a frequency distribution, various methods are available, including histograms, Bar Graphs, Frequency Polygons, and Pie Charts.

Types of frequency distribution graphs



5. Data Quality Issues

5.1 Missing Values

Missing values occur when no data is recorded for a variable. Common reasons include sensor errors, manual entry mistakes, or data corruption.

Missing values can:

- Reduce model accuracy
- Introduce bias
- Cause errors during training

Common handling techniques:

- Removing missing data
- Replacing with mean, median, or mode
- Using forward or backward fill methods

5.2 Outliers

What are Outliers?

Outliers are data points that lie far away from the majority of observations in a dataset. These values are unusually high or low when compared to the normal range of data.

An outlier does not always indicate an error; it may also represent a rare but important event.

Example of an Outlier

Consider a dataset representing monthly salaries (in ₹):

25,000 28,000 30,000 32,000 3,00,000

Here, 3,00,000 is an **outlier** because it is significantly larger than the other values.

Outliers can distort statistical measures like mean and standard deviation and negatively affect model performance.

Why Are Outliers Important in EDA?

Outliers significantly influence statistical measures and Machine Learning models.

Impact on Statistics

- Skews the mean
- Increases variance and standard deviation
- Misrepresents data distribution

Effect of Outliers on Mean and Median

Without Outlier: 10 12 11 13

Mean = 11.5, Median = 11.5

With Outlier: 10 12 11 13 100

Mean = 29.2, Median = 12

✓ Median remains stable

✗ Mean is highly affected

This is why median is preferred in the presence of outliers.

Common detection methods:

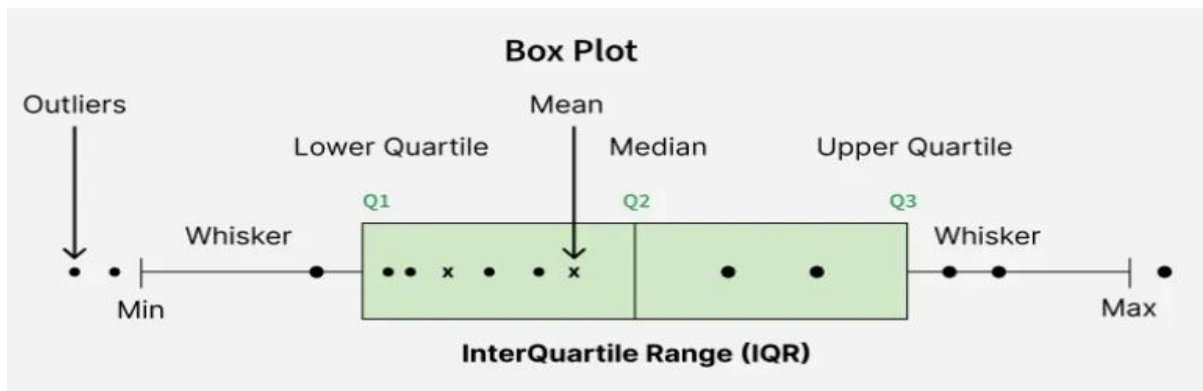
- Box plots
- Interquartile Range (IQR)
- Z-score method

a) Box Plot Method

Description:

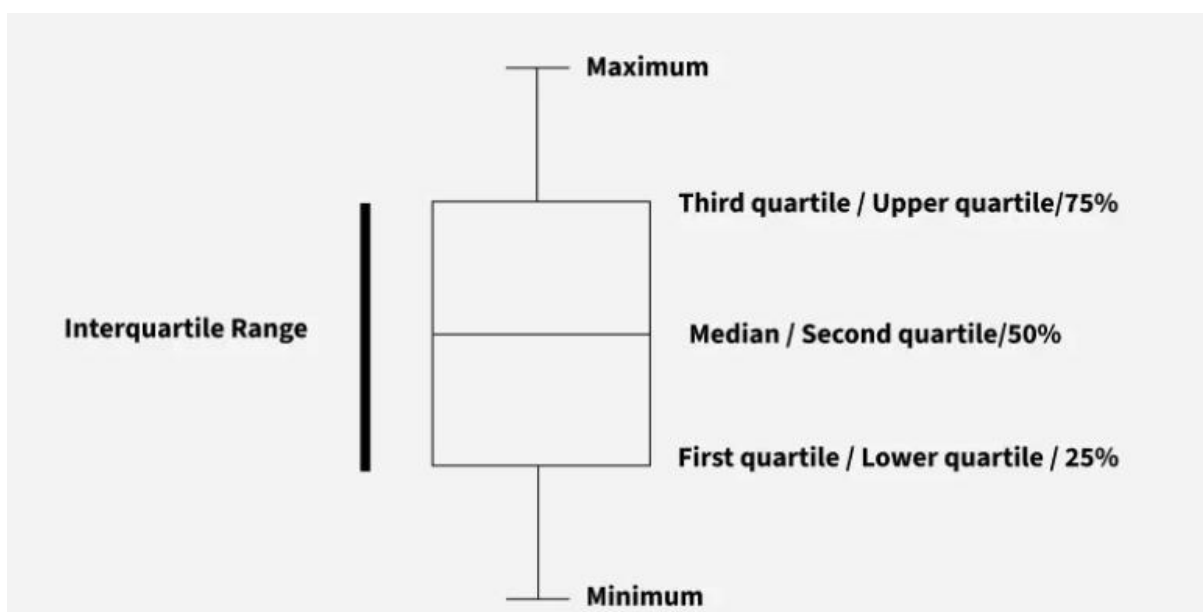
A **Box Plot** is a data visualization that summarizes a dataset's distribution. It shows key features like the range, median and spread of the data which makes it easier to understand the overall pattern.

A Box Plot consists of three main parts:



1. **Box:** The box spans from the first quartile (Q1) to the third quartile (Q3) representing the interquartile range (IQR). This box contains the middle 50% of the data with a line inside it representing the median (Q2).
2. **Whiskers:** The lines extending from the box called whiskers shows the range of the data. They extend from Q1 to the minimum value and from Q3 to the maximum value excluding outliers.
3. **Outliers:** Data points that fall outside the whiskers are considered outliers. These are marked separately and represent values that are significantly higher or lower than the rest of the data.

A box plot is a graphical visualization technique that represents the five-number summary of a dataset: minimum, first quartile (Q1), median, third quartile (Q3), and maximum. It provides a compact view of data distribution, central tendency, and variability.



1. **Minimum:** Smallest value in the dataset excluding outliers.
2. **First Quartile (Q1):** Middle value of the lower half of the data (25th percentile).
3. **Median (Q2):** Middle value that splits the dataset into two equal parts (50th percentile).
4. **Third Quartile (Q3):** Middle value of the upper half of the data (75th percentile).
5. **Maximum:** Largest value in the dataset excluding outliers.

Outlier Detection:

In a box plot, outliers are shown as individual points (often circles or dots) that lie beyond the upper or lower whiskers. These points indicate values that are significantly different from the rest of the data.

Usage:

- Best suited for quick visual inspection of datasets
- Useful for identifying skewness
- Effective for comparing distributions across multiple categories

b) Interquartile Range (IQR) Method

Description:

The Interquartile Range (IQR) measures data variability by calculating the difference between the 75th percentile (Q3) and the 25th percentile (Q1).

$$IQR = Q3 - Q1$$

Outlier Detection:

A data point is considered an outlier if it lies outside the following bounds:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

Values falling below the lower bound or above the upper bound are classified as outliers.

Usage:

- Robust to extreme values
- Works well for **skewed or non-normal data**
- Preferred when the dataset contains asymmetry or heavy tails

c) Z-Score Method

Description:

The Z-score method measures how many standard deviations (σ) a data point is away from the mean (μ). It standardizes data to identify extreme values.

Formula:

$$Z = \frac{(X - \mu)}{\sigma}$$

Outlier Detection:

A data point is flagged as an outlier if its **absolute Z-score** exceeds a predefined threshold, typically:

$$|Z| > 3$$

Usage:

- Effective for datasets that follow a **Gaussian (normal) distribution**
- Commonly used in statistical analysis
- Less reliable for skewed data or datasets with heavy outliers

Comparison Summary

Method	Basis	Best Used For
Box Plot	Quartiles (visual)	Quick inspection & comparison
IQR	Quartile range	Skewed or non-normal data

Method	Basis	Best Used For
Z-Score	Mean & std deviation	Normally distributed data

6. Data Visualization Techniques

Data visualization helps in understanding data patterns through graphical representation.

What is Data Visualization?

- Turning data into visual formats like charts, graphs, and maps
- Makes complex data easy to understand at a glance
- Helps identify trends, patterns, and insights



Benefits of Data Visualization

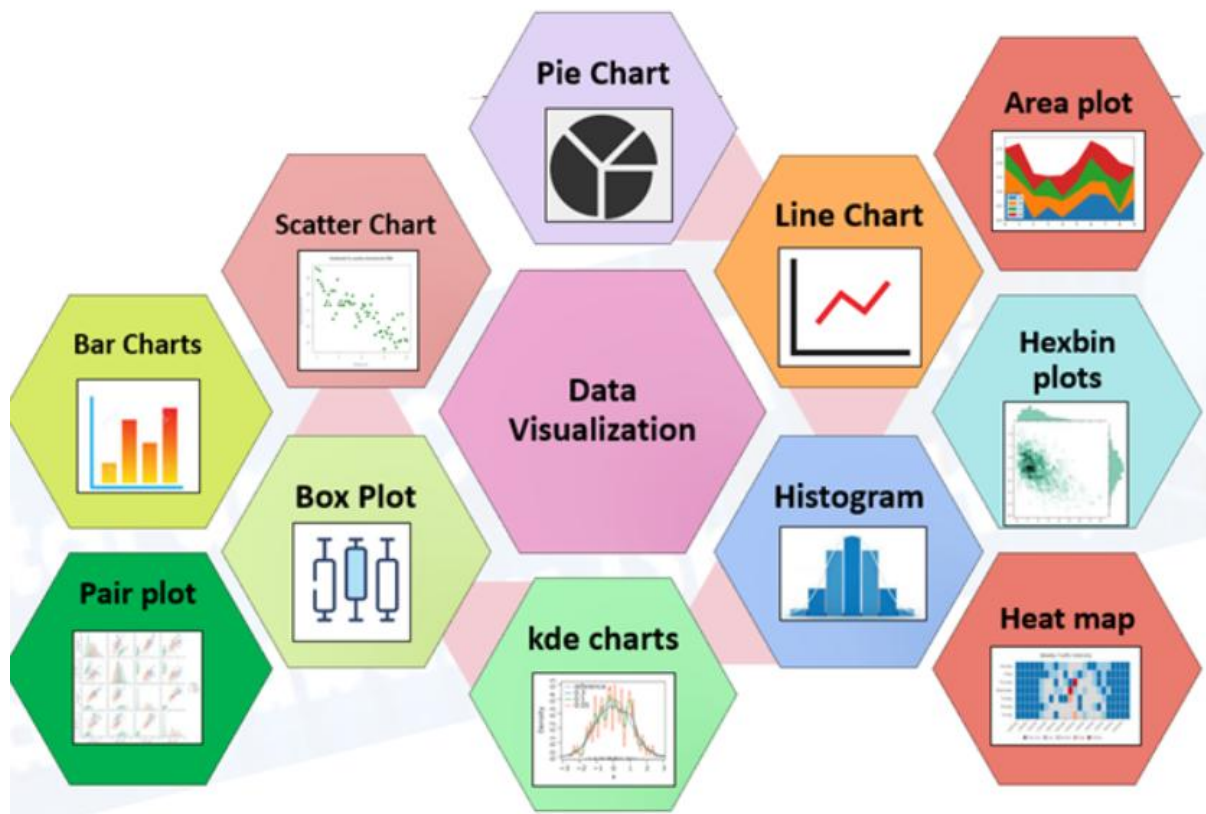
- Faster Insights**: Represented by an icon of a lightbulb and a bar chart.
- Better Decision-Making**: Represented by an icon of a person with a checkmark and a cross.
- Simplifies Complex Data**: Represented by an icon of a branching arrow.
- Improved Communication**: Represented by an icon of two speech bubbles.
- Time Efficiency**: Represented by an icon of a clock and a checkmark.

Common visualization techniques:

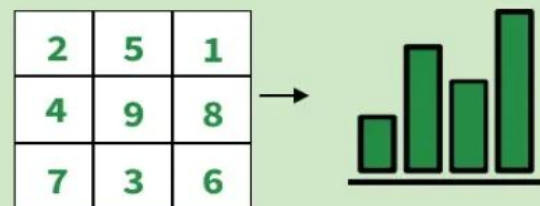
- **Histogram:** Distribution of numerical data
- **Bar Chart:** Comparison of categories

- **Scatter Plot:** Relationship between two variables
- **Box Plot:** Detection of outliers

Visualization makes data insights intuitive and easier to interpret.



- Matplotlib is a Python library.
- It is used for creating static, animated and interactive data visualizations.



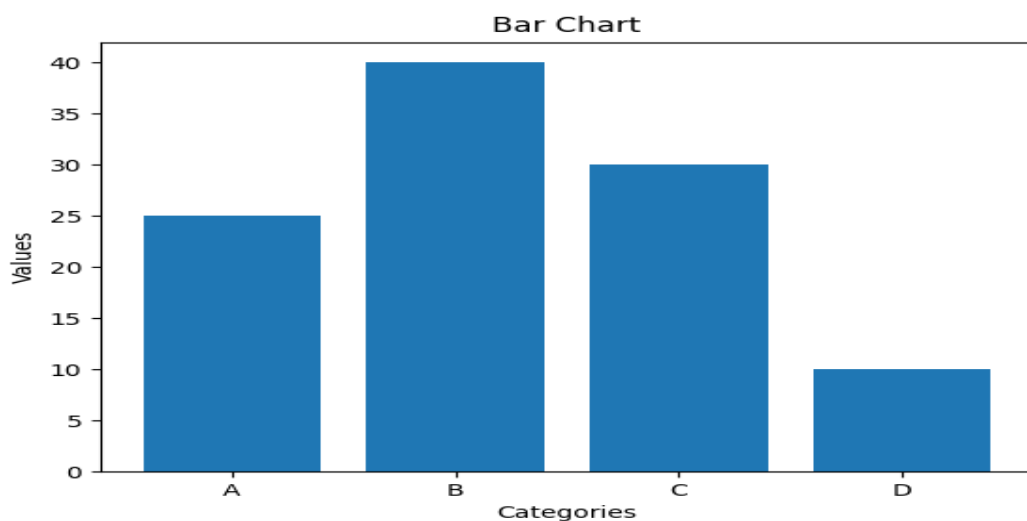
- Seaborn is a Python library built on Matplotlib.
- It helps in creating statistical visualizations.
- It works seamlessly with Pandas and provides attractive plots.



1. Bar Charts

Bar charts are used to compare values across different categories using rectangular bars. X-axis shows categories while Y-axis represents values. Common types include horizontal, stacked and grouped bar charts.

Below is the Example of Bar Chart:



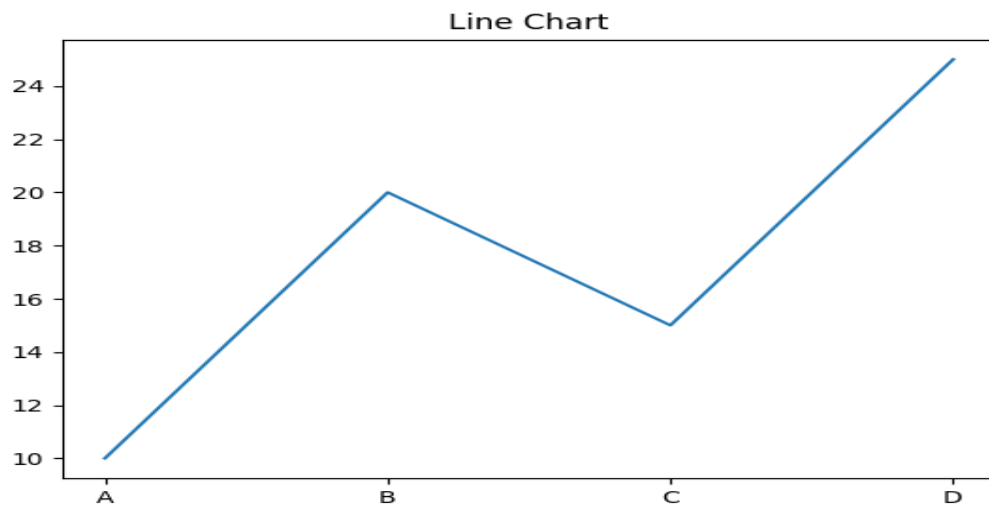
When to Use:

- To compare different categories
- To rank values from highest to lowest
- To show relationships between multiple variables

2. Line Charts

Line charts show how values change over time by connecting data points with lines. They help visualize trends like increases, decreases or stability.

Below is the example of line chart:



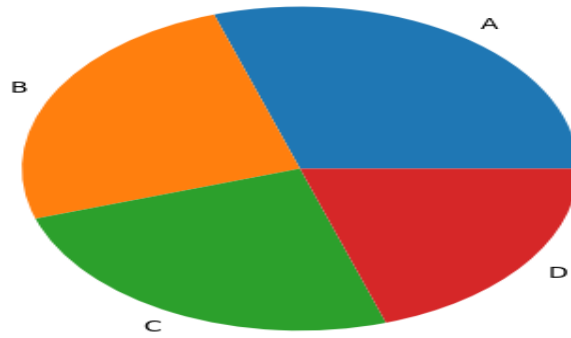
When to Use:

- To track changes over time
- To compare trends across multiple data series
- For time series analysis

3. Pie Charts

Pie charts are round charts divided into slices, where each slice shows a part of the whole. The size of each slice represents its percentage.

Below is the example of pie chart:



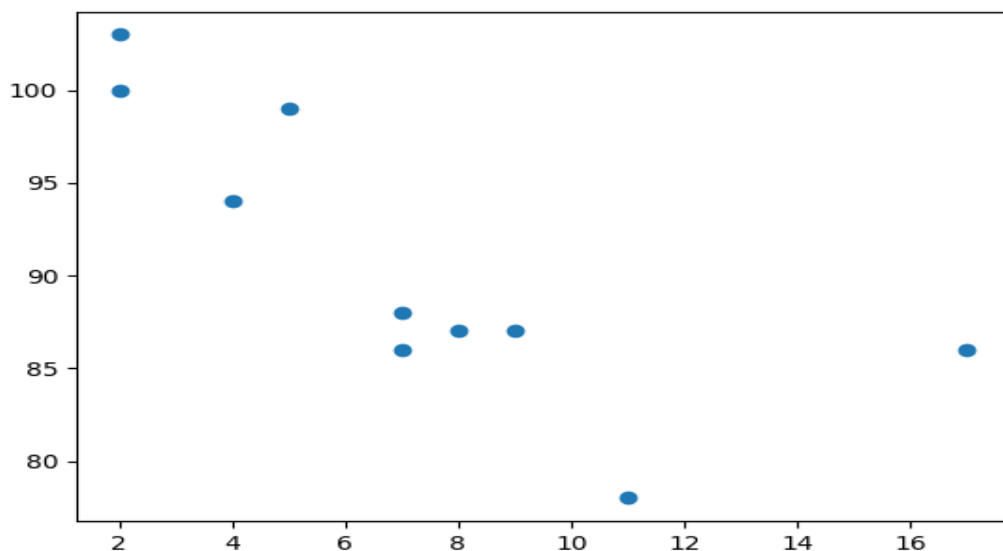
When to Use:

- To show how different parts contribute to a whole
- To highlight a dominant category

4. Scatter Chart (Plots)

Scatter charts use dots to show relationship between two numerical variables. X-axis shows the independent variable and Y-axis shows the dependent variable.

Below is the example of scatter chart:



When to Use:

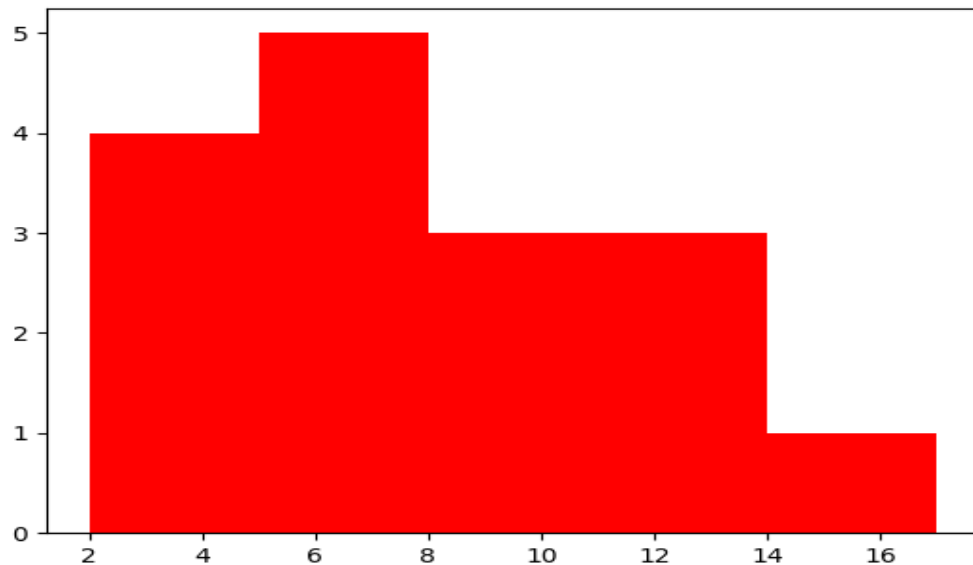
- To observe relationships between two variables

- To detect patterns, clusters or outliers in data

5. Histogram

A histogram displays the distribution of numerical data by grouping values into intervals (bins) and showing their frequency as bars. It helps reveal the shape, spread and patterns in the data.

Below is the example of histogram:



Representation of Histogram

When to Use:

- To visualize the distribution of numerical data
- To explore patterns, trends and outliers

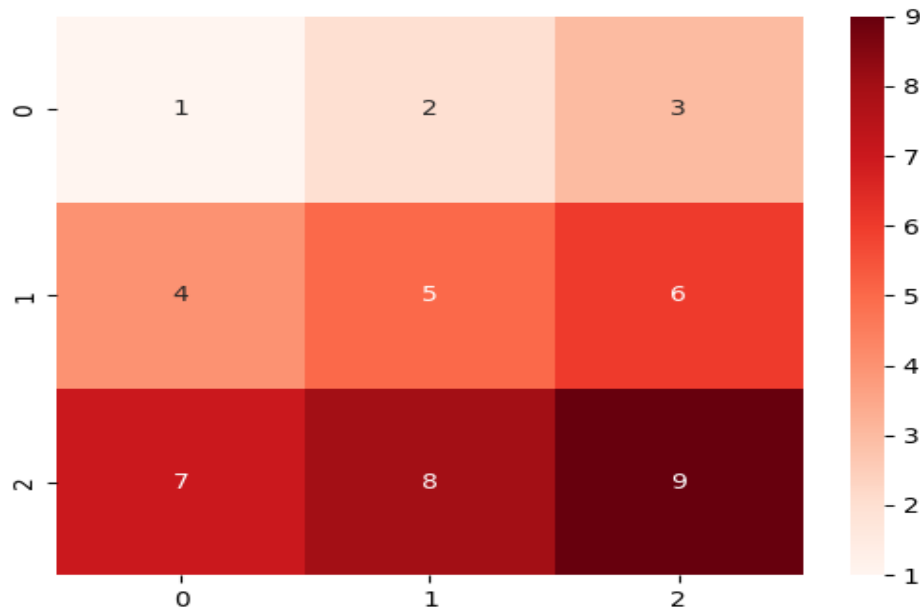
Advanced Charts for Data Visualization

Advanced charts are designed to handle more complex data. They help analyze multiple variables, uncover deeper insights and reveal patterns that might be missed with basic visuals.

1. Heatmap

A heatmap displays data in a matrix format using color to represent values. It's ideal for spotting patterns, correlations and variations in large datasets.

Below is the example of heatmap:



Representation of Heatmap

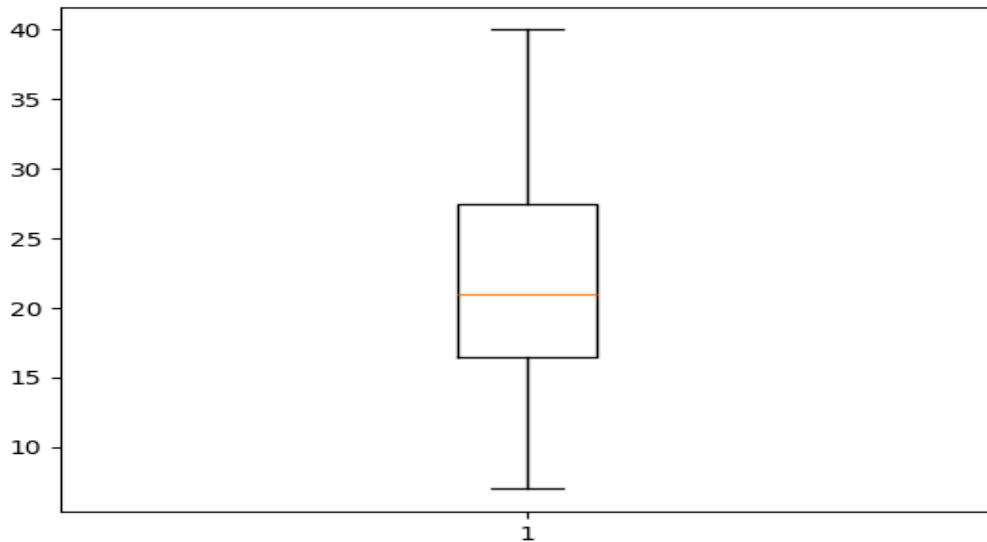
When to Use:

- To identify clusters or groupings in data
- To visualize correlations between variables
- For risk analysis in fields like finance or network security

2. Box Plot (Box-and-Whisker Plot)

A box plot displays distribution of numerical data, showing median, quartiles and outliers. It's useful for understanding variability and detecting unusual values.

Below is the example of box plot:



Representation of Box plot

When to Use:

- To detect outliers in data
- To compare distributions across groups
- To visualize spread and variability

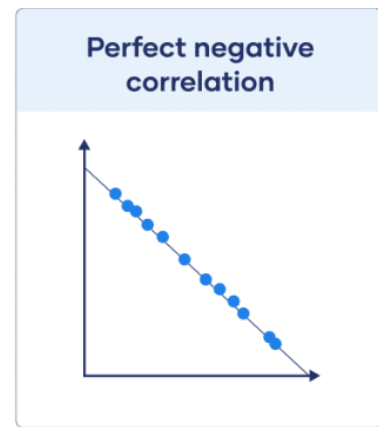
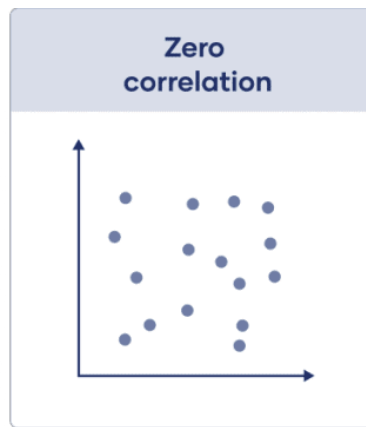
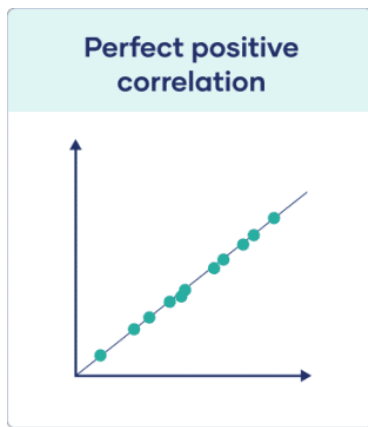
7. Correlation Analysis

Correlation measures the relationship between two numerical variables.

- **Positive Correlation:** Both variables increase together
- **Negative Correlation:** One increases while the other decreases
- **Zero Correlation:** No relationship

Correlation values range from **-1 to +1**.

A correlation analysis is a statistical technique used to determine whether a relationship exists between two or more variables. In other words, it helps to understand if one variable moves as a function of another, if they are related in some way.



a) Positive Correlation

Definition:

A positive correlation exists when **both variables increase or decrease together**.

- As one variable increases, the other also increases
- As one variable decreases, the other also decreases

Correlation Range:

$$0 < r \leq +1$$

Example:

- Height and weight
- Years of experience and salary
- Study hours and exam marks

Interpretation:

If study hours increase, exam scores tend to increase.

b) Negative Correlation

Definition:

A negative correlation exists when **one variable increases while the other decreases**.

- Increase in one variable leads to a decrease in the other

Correlation Range:

$$-1 \leq r < 0$$

Example:

- Price and demand
- Speed and travel time
- Temperature and heater usage

Interpretation:

As temperature increases, heater usage decreases.

c) Zero (No) Correlation**Definition:**

Zero correlation indicates **no linear relationship** between variables.

- Change in one variable does not affect the other

Correlation Value:

$$r \approx 0$$

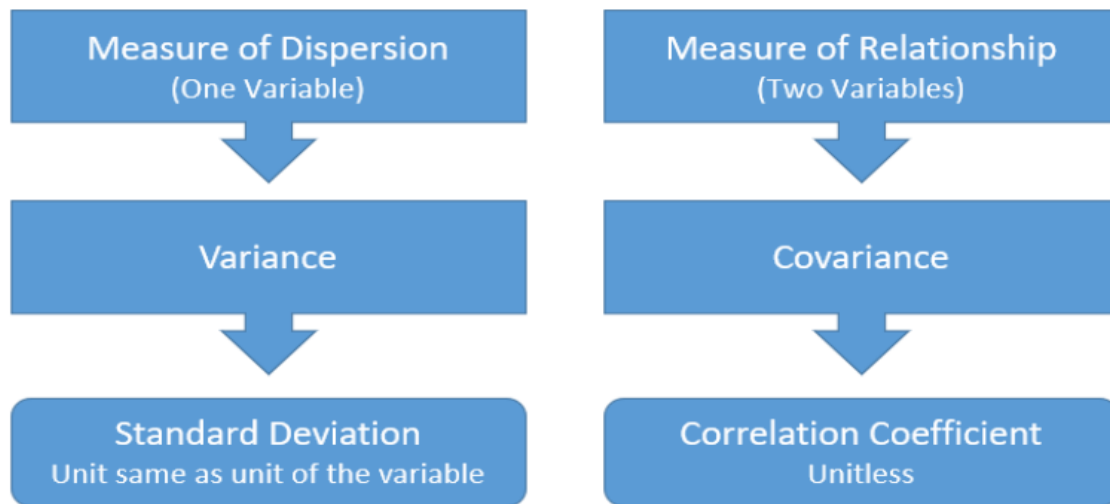
Example:

- Shoe size and intelligence
- Roll number and exam marks

Interpretation:

Variables are independent of each other.

Covariance & Correlation



7.2 Causation

Causation refers to a cause-and-effect relationship, where a change in one variable directly produces a change in another variable.

If variable X causes Y, then modifying X will result in a change in Y, assuming other factors remain constant.

Examples:

- Rain causes wet roads
- Fire causes smoke
- Exercise contributes to weight loss

8. Tools for EDA in Python

Python provides powerful libraries for EDA:

- **NumPy:** Numerical computations
- **Pandas:** Data manipulation and analysis
- **Matplotlib:** Basic visualization
- **Seaborn:** Statistical data visualization

These tools simplify data analysis and help in faster insight generation.

9. Practical EDA Workflow

A typical EDA workflow includes:

1. Loading the dataset
2. Inspecting structure and data types
3. Computing descriptive statistics
4. Handling missing values
5. Detecting outliers
6. Visualizing data
7. Performing correlation analysis

10. Summary

Exploratory Data Analysis is a fundamental step in Machine Learning that helps in understanding and preparing data for modeling. By using descriptive statistics, visualization techniques, and correlation analysis, EDA enables better decision-making and improves model reliability. A well-performed EDA ensures that Machine Learning models are built on clean, meaningful, and unbiased data.

Q&A

QUESTION 1:

A dataset shows a very high average value, while most observations are concentrated at the lower end. Which measure would best describe the central tendency of this data and why?

ANSWER:

The median would best describe the central tendency because it is not influenced by extreme values. In skewed datasets, the median provides a more realistic representation of the typical value than the mean.

QUESTION 2:

A machine learning model performs poorly despite having a large amount of data. Later analysis reveals missing and inconsistent values. Which analytical step could have helped avoid this issue?

ANSWER:

Exploratory Data Analysis could have helped by identifying missing values, inconsistencies, and data quality issues before model training, ensuring cleaner and more reliable input data.

QUESTION 3:

An analyst wants to understand how numerical values are distributed across different ranges within a dataset. Which method should be used to achieve this?

ANSWER:

A frequency distribution, commonly visualized using a histogram, should be used to observe how values are spread across different intervals and to identify patterns or skewness.

QUESTION 4:

While analyzing salary data, one extremely high value significantly increases the average. How does this affect statistical interpretation?

ANSWER:

The presence of an outlier distorts the mean and increases measures of variability, leading to a misleading interpretation of the data's overall distribution.

QUESTION 5:

A dataset contains skewed values and does not follow a normal distribution. Which outlier detection technique is most suitable and why?

ANSWER:

The Interquartile Range (IQR) method is most suitable because it relies on quartiles rather than mean and standard deviation, making it effective for skewed and non-normal data.

QUESTION 6:

Two variables tend to increase together, but changing one does not directly result in a change in the other. How should this relationship be understood?

ANSWER:

This relationship should be understood as positive correlation rather than causation, indicating association without a direct cause-and-effect link.

QUESTION 7:

Eliminating records with missing values leads to a significant reduction in dataset size. What alternative approach should be adopted?

ANSWER:

Missing values should be handled using appropriate imputation techniques such as mean, median, or mode replacement.

QUESTION 8:

An analyst needs to compare multiple groups of numerical data and identify unusual values efficiently. Which visualization technique is most effective?

ANSWER:

A box plot is most effective as it displays median, quartiles, data spread, and outliers for multiple groups in a single visualization.

QUESTION 9:

Before training a model, an analyst wants to detect redundant features among numerical variables. Which EDA technique should be applied?

ANSWER:

Correlation analysis should be applied to identify strong relationships between variables and reduce redundancy during feature selection.

QUESTION 10:

A model performs well on training data but poorly on unseen data. How could early data analysis have helped reduce this issue?

ANSWER:

Early data analysis could have identified skewed distributions, outliers, or highly correlated features, enabling better preprocessing and feature selection to reduce overfitting.